

## **Task - 15: Vulnerability Assessment & Risk Prioritization**

### **Internship Task – Elevate Labs**

This task focused on performing a vulnerability assessment and prioritizing security risks based on their impact. The objective was to identify weaknesses in systems and classify them according to severity for effective remediation.

#### **1. Defining Scope**

The first step was to define the scope and target systems for assessment. This helps ensure that the scanning process remains focused and relevant.

#### **2. Using Vulnerability Scanning Tools**

Tools such as Nessus Essentials or OpenVAS were used to perform the scan. These tools help identify known vulnerabilities in systems and applications.

#### **3. Running the Scan**

A vulnerability scan was executed and monitored to detect potential weaknesses. The scan identified outdated software, misconfigurations, and security gaps.

#### **4. Reviewing Vulnerabilities**

Each identified vulnerability was reviewed carefully. Severity levels were analyzed to understand potential impact.

#### **5. CVE Mapping**

Detected vulnerabilities were mapped to CVE (Common Vulnerabilities and Exposures). This helps in understanding the nature of known security flaws.

#### **6. CVSS Scoring**

CVSS scores were used to evaluate the severity of vulnerabilities. This provides a standardized way to measure risk.

#### **7. Risk Classification**

Vulnerabilities were classified into categories such as: - High Risk - Medium Risk - Low Risk

This classification helps in prioritization.

#### **8. Risk Prioritization**

Critical vulnerabilities were prioritized for immediate remediation. Lower risk issues were scheduled for later resolution.

## **9. Remediation Suggestions**

Recommended actions included: - Applying security patches - Updating software - Fixing misconfigurations

### **Learning Outcome**

Through this task, I gained practical knowledge of vulnerability assessment. I learned how to identify system weaknesses, evaluate risk levels, and prioritize remediation efforts effectively.